Hints: Linear algebra – 12. Mixed problems: commutators, traces and the numerical range

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $M = \begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix}$ has rank 1 and eigenvalues 0 and 8. Any solution X commutes with $M = X^n$.

Hint 2. In an eigenbasis of M (distinct eigenvalues), X is diagonal, $\operatorname{diag}(a, b)$ with $a^n = 0$ and $b^n = 8$. So the solutions are $X = \frac{b}{8}M$ with $b = 8^{1/n}\omega$, $\omega^n = 1$, and $\operatorname{tr} X = b$. The sum of the n-th roots of unity is 0 for $n \geqslant 2$.

Problem 2

Hint 1. For skew-symmetric A, $A^2 = -A^TA$. So $A^2 + B^2 + C^2 = -(A^TA + B^TB + C^TC)$ is negative semidefinite.

Hint 2. Singularity gives $x \neq 0$ with $\|Ax\|^2 + \|Bx\|^2 + \|Cx\|^2 = 0$, so $Ax = Bx = Cx = 0$. Then $x^TA = -(Ax)^T = 0$, and similarly for B and C, so $x^T(AP + BQ + CR) = 0$.

Problem 3

Hint 1. Put $s = u + v > 0$ and $w = u - v$. Then $As = s$ and $Aw = -w$. You want to show $w = 0$ (Perron–Frobenius would do it; here is an elementary route).

Hint 2. Let $c = \max_i |w_i|/s_i$, attained at i. Then $cs_i = |w_i| \leqslant \sum_j a_{ij}|w_j| \leqslant c\sum_j a_{ij}s_j = cs_i$. Since all $a_{ij} > 0$, equality forces $w = \pm cs$, and then $Aw = w$. Together with $Aw = -w$ this gives $w = 0$.

Problem 4

Hint 1. Expand $(A+B)^3 - A^3 - B^3$ into six words. If $AB = O$, then $BA = (AB)^T = O$ and all six words vanish.

Hint 2. Conversely, take the trace: $3\operatorname{tr}(A^2B) + 3\operatorname{tr}(AB^2) = 0$. Here $\operatorname{tr}(A^2B) = \|B^{1/2}A\|_F^2 \geqslant 0$, and similarly $\operatorname{tr}(AB^2) \geqslant 0$. So $B^{1/2}A = O$, hence $BA = O$ and $AB = (BA)^T = O$.

Problem 5

Hint 1. Group the right-hand side with $C = AB - BA$: it is $C + (AC - CA) + ACA$.

Hint 2. Suppose A is invertible, multiply by A^{-1} and take traces. You get $\operatorname{tr}(A^{-1}C) = \operatorname{tr}(B - A^{-1}BA) = 0$, $\operatorname{tr} C = 0$, $\operatorname{tr}(A^{-1}CA) = 0$ and $\operatorname{tr}(CA) = \operatorname{tr}(ABA - BA^2) = 0$. This gives $n = \operatorname{tr} E = 0$, a contradiction.

Problem 6

Hint 1. Over $\mathbb{C}$, A is diagonalisable with eigenvalues $\pm i$. Its eigenspaces V_+ and $V_- = \overline{V_+}$ have the same dimension, and B preserves both because $AB = BA$.

Hint 2. $\det B = \det(B|_{V_+})\det(B|_{V_-})$. Since B is real, the second factor is the conjugate of the first, so $\det B = |\det(B|_{V_+})|^2 \geqslant 0$.

Problem 7

Hint 1. Compare the operator norm with the Frobenius norm: $\|A\|_2 \leqslant \|A\|_F < 1/\sqrt{5} < 1$. Then -1 cannot be an eigenvalue of A. (Alternatively, bound $|a+d| + |ad-bc|$ in $\det(E+A) = 1 + a + d + ad - bc$.)

Problem 8

Hint 1. For an idempotent, $N(A_i) = \operatorname{rank}(E - A_i)$.

Hint 2. Telescope: $E - A_1 \cdots A_k = \sum_{i=1}^{k} A_1 \cdots A_{i-1}(E - A_i)$. Then use subadditivity of rank and $\operatorname{rank}(XY) \leqslant \operatorname{rank} Y$.

Problem 9

Hint 1. Try U a permutation matrix, $Ue_j = e_{\sigma(j)}$. Then $AU - UA$ sends e_j to $(\lambda_{\sigma(j)} - \lambda_j)e_{\sigma(j)}$, so $|AU - UA| = \operatorname{diag}(|\lambda_{\sigma(j)} - \lambda_j|)$. Also $|A - \lambda E| = \operatorname{diag}(|\lambda_j - \lambda|)$.

Hint 2. Sort $\lambda_1 \leqslant \ldots \leqslant \lambda_n$, take $\sigma(j) = n + 1 - j$ and let λ be a median. Then λ lies between λ_j and λ_{n+1-j} for every j, so $|\lambda_{n+1-j} - \lambda_j| \geqslant |\lambda_j - \lambda|$.

Problem 10

(a)

Hint 1. The hypotheses say $(A + iB)(C + iD) = E$. A one-sided inverse of a square matrix is two-sided, so $(C + iD)(A + iB) = E$. Separate real and imaginary parts.

(b)

Hint 1. Let $M = A + iB$, so $M^{-1} = C + iD$, and set $L = M^{-1}\bar{M}$. Then $\bar{L} = L^{-1}$, and $A = \frac{1}{2}M(E + L)$, $C = \frac{1}{2}(E + L^{-1})M^{-1}$ (use $\bar{M}^{-1} = L^{-1}M^{-1}$).

Hint 2. $\det(AC) = 4^{-n}\det(E + L)\det(E + \bar{L}) = 4^{-n}|\det(E + L)|^2$. Similarly $B = \frac{1}{2i}M(E - L)$ and $D = \frac{1}{2i}(E - L^{-1})M^{-1}$. These give $\det(BD) = (-4)^{-n}|\det(E - L)|^2$.

Problem 11

Hint 1. The hypothesis says $[A, [A, B]] = O$: $C = AB - BA$ commutes with A.

Hint 2. Show $\operatorname{tr} C^k = 0$ for all $k \geqslant 1$. Write $C^k = C^{k-1}AB - C^{k-1}BA$ and use $C^{k-1}A = AC^{k-1}$ and the cyclicity of the trace. If all power sums of the eigenvalues vanish, the matrix is nilpotent.

Problem 12

Hint 1. The commutator $B^2 = AB - BA$ commutes with B. As in the previous problem, $\operatorname{tr}(B^{2k}) = \operatorname{tr}\big(B^{2k-2}(AB - BA)\big) = 0$ for all $k \geqslant 1$.

Hint 2. So B^2 is nilpotent, hence B is nilpotent. For 2×2 this means $B^2 = O$, i.e. $AB - BA = O$.

Problem 13

Hint 1. $M = AB - BA + E$ has rank one, so $M^2 = (\operatorname{tr} M)M = nM$.

Hint 2. With $C = AB - BA$ this gives $C^2 = (n-2)C + (n-1)E$, so $\operatorname{tr} C^2 = n(n-1)$. Expanding, $\operatorname{tr} C^2 = 2\operatorname{tr}(ABAB) - 2\operatorname{tr}(A^2B^2)$.

Problem 14

Hint 1. Look for a right inverse of $A - B$. Note $(A-B)A^{-1} = E - BA^{-1}$.

Hint 2. So $(A-B)(C+A^{-1}) = E$, hence $(C+A^{-1})(A-B) = E$. Expand.

Problem 15

Hint 1. It suffices to show that the segment between two points $\langle Ax, x\rangle$ and $\langle Ay, y\rangle$ lies in $W(A)$. Replacing A by $aA+bE$, assume the points are 0 and 1. Write $A = H + iK$ with H, K Hermitian; then $\langle Kx, x\rangle = \langle Ky, y\rangle = 0$.

Hint 2. Replace y by $e^{i\theta}y$ so that $\operatorname{Re}\langle Kx, y\rangle = 0$. Then $\langle Kz, z\rangle = 0$ for all $z = x + sy$, $s \in \mathbb{R}$, so $\langle Az, z\rangle / \|z\|^2$ is real. It runs continuously from 0 ($s = 0$) to 1 ($s \to \infty$).

Hint 3. Real version: the joint range of $x \mapsto (x^TAx, x^TBx)$ on the sphere is convex if its intersection with every line is connected. Show that the preimage of a line, $\{x : \|x\| = 1,\ x^TQx = 0\}$ with $Q = \alpha A + \beta B - \gamma E$, is connected or consists of two antipodal pieces when $n \geqslant 3$ (diagonalise Q). Since x and $-x$ have the same image, the intersection is connected.

Problem 16

Hint 1. Let $C = AB - BA$ and compute $\operatorname{tr} C^2 = 2\operatorname{tr}\big((AB)^2\big) - 2\operatorname{tr}(A^2B^2) = 0$. Also $\operatorname{tr} C = 0$.

(a)

Hint 1. The eigenvalues of C are real and $\sum \lambda_i^2 = 0$, so all of them are 0.

(b)

Hint 1. For $n = 2$ the eigenvalues are $\lambda, -\lambda$ and $2\lambda^2 = 0$, whether λ is real or not. For $n \geqslant 3$ you need a real matrix C with $\operatorname{tr} C = \operatorname{tr} C^2 = 0$ that is not nilpotent, and you must write it as a commutator.

Hint 2. Take C the permutation matrix of a 3-cycle (characteristic polynomial $x^3 - 1$), padded with zeros. A matrix with zero diagonal is a commutator: with $A = \operatorname{diag}(1, 2, \ldots, n)$ and $b_{ij} = c_{ij}/(i-j)$ for $i \neq j$, $b_{ii} = 0$, one gets $AB - BA = C$.